Part 1 : Library Management System – Book Lifecycle Modeling

1. Introduction

In a library, a book goes through several states during its lifecycle. Properly modeling these states allows the library to manage borrowing, reservations, maintenance, and loss of books. This document focuses on how a book's state changes in the book life cycle, and how to handle concurrent requests safely.

Book States:

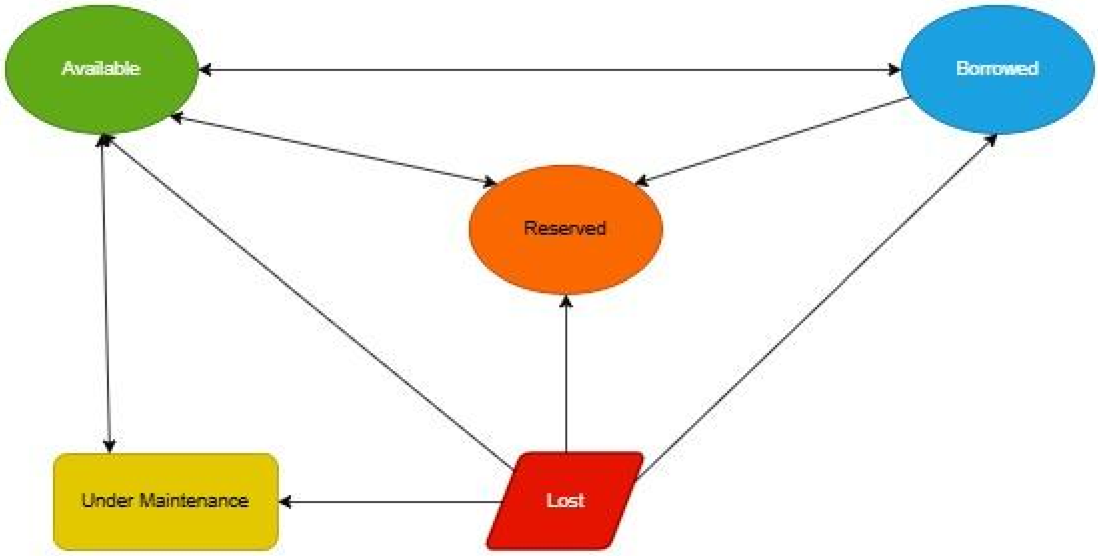
- Available : Book is in the library and can be borrowed or reserved or under Maintenance or lost.
- Borrowed : Book has been borrowed by a member.
- Reserved : Book has been reserved for a member.
- Lost : Book is reported lost and cannot be borrowed / reserved.
- Under Maintenance – Book is temporarily unavailable due to maintenance or repair.

2. Book State Lifecycle

2.1 State Transitions Table

Current State	Event / Action	Next State
Available	Borrow	Borrowed
Available	Reserve	Reserved
Available	Send for Maintenance	Under Maintenance
Available	Report Lost	Lost
Reserved	Borrow (by reserving member)	Borrowed
Reserved	Cancel Reservation	Available
Reserved	Report Lost	Lost
Borrowed	Return	Available
Under Maintenance	Maintenance Complete	Available
Under Maintenance	Report Lost	Lost

2.2 Book State Lifecycle :



3. Handling Concurrency in book state

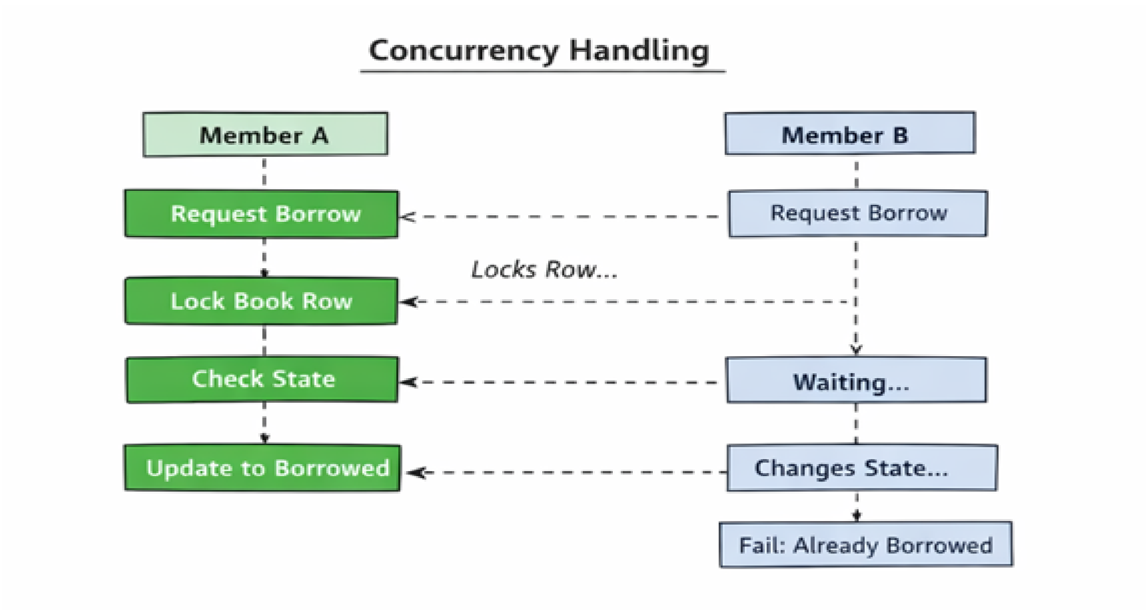
3.1 Problem

- Multiple members can try to borrow or reserve the same book at the same time.
- Without proper locking, the book state can become inconsistent (race condition).

3.2 Solution

- **Pessimistic locking** ensures only one transaction modifies a book at a time.
- Use **row-level locks** with database transactions.

3.3 Concurrency Handling workflow Diagram :



4. Summary

- The book lifecycle model ensures proper tracking of availability, reservations, maintenance, and loss.
- State transitions and diagrams clearly define possible actions.
- Concurrency is handled using pessimistic locking, preventing race conditions during simultaneous borrow/reserve requests.
- This model is robust, consistent, and ready for implementation in a library management system.